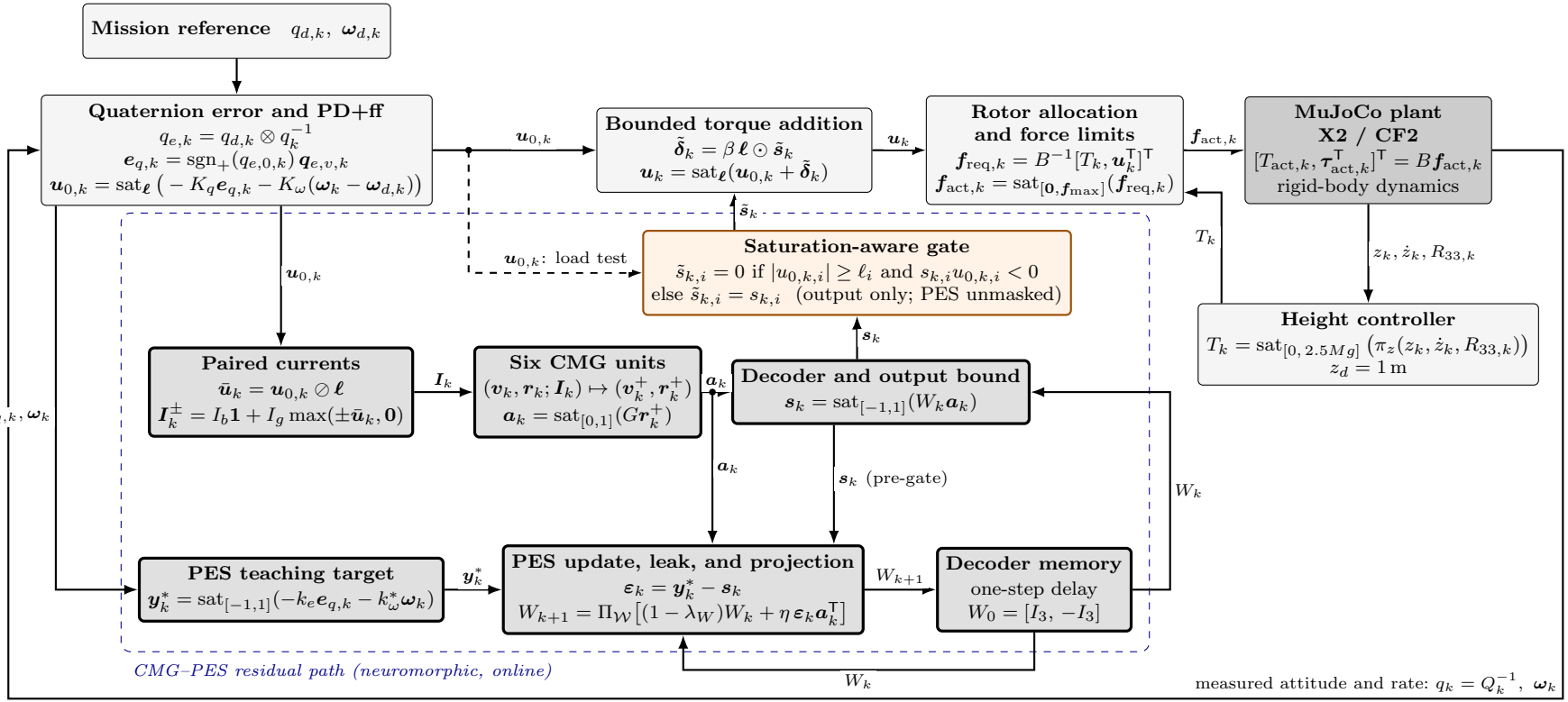

Mission reference $q_{d,k},\ \boldsymbol{\omega}_{d,k}$
Quaternion error and PD+ff
$q_{e,k} = q_{d,k} \otimes q_k^{-1}$
$\boldsymbol{e}_{q,k} = \mathrm{sgn}_+(q_{e,0,k})\,\boldsymbol{q}_{e,v,k}$
$\boldsymbol{u}_{0,k} = \mathrm{sat}_{\boldsymbol{\ell}}\left(-K_q\boldsymbol{e}_{q,k} - K_\omega(\boldsymbol{\omega}_k - \boldsymbol{\omega}_{d,k})\right)$
$\boldsymbol{u}_{0,k}$
Bounded torque addition
$\tilde{\boldsymbol{\delta}}_k = \beta\,\boldsymbol{\ell} \odot \tilde{\boldsymbol{s}}_k$
$\boldsymbol{u}_k = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_{0,k} + \tilde{\boldsymbol{\delta}}_k)$
$\boldsymbol{u}_k$
Rotor allocation and force limits
$\boldsymbol{f}_{\mathrm{req},k} = B^{-1}[T_k, \boldsymbol{u}_k^{\mathsf{T}}]^{\mathsf{T}}$
$\boldsymbol{f}_{\mathrm{act},k} = \mathrm{sat}_{[\boldsymbol{0}, \boldsymbol{f}_{\max}]}(\boldsymbol{f}_{\mathrm{req},k})$
$\boldsymbol{f}_{\mathrm{act},k}$
MuJoCo plant
X2 / CF2
$[T_{\mathrm{act},k}, \boldsymbol{\tau}_{\mathrm{act},k}^{\mathsf{T}}]^{\mathsf{T}} = B\boldsymbol{f}_{\mathrm{act},k}$
rigid-body dynamics
$z_k, \dot{z}_k, R_{33,k}$
Height controller
$T_k = \mathrm{sat}_{[0,\ 2.5Mg]}\left(\pi_z(z_k, \dot{z}_k, R_{33,k})\right)$
$z_d = 1\,\mathrm{m}$
T_k
$\tilde{\boldsymbol{s}}_k$
$\boldsymbol{u}_{0,k}$: load test
Saturation-aware gate
$\tilde{s}_{k,i} = 0$ if $|u_{0,k,i}| \geq \ell_i$ and $s_{k,i}u_{0,k,i} < 0$
else $\tilde{s}_{k,i} = s_{k,i}$ (output only; PES unmasked)
$\boldsymbol{s}_k$
$\boldsymbol{u}_{0,k}$
Paired currents
$\bar{\boldsymbol{u}}_k = \boldsymbol{u}_{0,k} \oslash \boldsymbol{\ell}$
$\boldsymbol{I}_k^{\pm} = I_b\mathbf{1} + I_g\max(\pm\bar{\boldsymbol{u}}_k, \boldsymbol{0})$
$\boldsymbol{I}_k$
Six CMG units
$(\boldsymbol{v}_k, \boldsymbol{r}_k; \boldsymbol{I}_k) \mapsto (\boldsymbol{v}_k^+, \boldsymbol{r}_k^+)$
$\boldsymbol{a}_k = \mathrm{sat}_{[0,1]}(G\boldsymbol{r}_k^+)$
$\boldsymbol{a}_k$
Decoder and output bound
$\boldsymbol{s}_k = \mathrm{sat}_{[-1,1]}(W_k\boldsymbol{a}_k)$
$\boldsymbol{a}_k$
$\boldsymbol{s}_k$ (pre-gate)
W_k
$\boldsymbol{e}_{q,k}, \boldsymbol{\omega}_k$
PES teaching target
$\boldsymbol{y}_k^* = \mathrm{sat}_{[-1,1]}(-k_e\boldsymbol{e}_{q,k} - k_\omega^*\boldsymbol{\omega}_k)$
$\boldsymbol{y}_k^*$
PES update, leak, and projection
$\boldsymbol{\varepsilon}_k = \boldsymbol{y}_k^* - \boldsymbol{s}_k$
$W_{k+1} = \Pi_{\mathcal{W}}\left[(1-\lambda_W)W_k + \eta\,\boldsymbol{\varepsilon}_k\boldsymbol{a}_k^{\mathsf{T}}\right]$
W_{k+1}
Decoder memory
one-step delay
$W_0 = [I_3,\ -I_3]$
W_k
CMG–PES residual path (neuromorphic, online)
measured attitude and rate: $q_k = Q_k^{-1},\ \boldsymbol{\omega}_k$